Frontend Theory Assignment - 1

Q1. HTTP Request-Response Cycle

The HTTP cycle is how a browser and server communicate. First, the client (browser) sends an HTTP Request over the internet (like clicking a link or typing a URL). The web server receives this request, processes it, and fetches the necessary data. Finally, the server sends back an HTTP Response containing the requested resources (like an HTML file), which the browser then displays.

Q2. Introduction to Web Technologies

HTML gives the basic structure to the web page. CSS is used to style the page, managing layouts and colors. JavaScript adds logic and interactivity to the page. Bootstrap is a CSS framework that helps build responsive designs quickly. jQuery is a JavaScript library that simplifies DOM manipulation. React.js is a library for building reusable user interface components.

Q3. Client-Side vs. Server-Side Technologies

Client-side technologies (like HTML, CSS, JS) run directly in the user's web browser to display the interface. Server-side technologies (like Node.js, Python, PHP) run on the web server in the background, securely handling databases, authentication, and application logic.

Q4. Block-Level Elements in HTML

Block-level elements always start on a new line and stretch out to take up the full horizontal width available on the page. Three common examples are <div>, <p> (paragraphs), and <h1> (headings).

Q5. Semantic HTML Elements

Semantic elements have tags that clearly describe their meaning and content to both the developer and the browser, such as <header>, <article>, or <footer>. They are important because they improve web accessibility for screen readers and help with search engine optimization (SEO).

Q6. Creating a Table Using HTML

```
<table border="1">

 <tr>

  <th>Sr. No.</th>
```

Done by:
Nikhilesh Yadav

```
   <th>Name</th>
   <th>Designation</th>
   <th>Department</th>
  </tr>
  <tr>
   <td>1</td>
     <td>John</td>
       <td>Manager</td>
       <td>Sales</td>
   </tr>
  <tr>
   <td>2</td>
   <td>Smith</td>
   <td>Sr. Manager</td>
   <td>Marketing</td>
   </tr>
  <tr>
   <td>3</td>
   <td>Jane</td>
   <td>HR</td>
   <td>Manager</td>
   </tr>
</table>
```

Q7. Creating a Nested List Using HTML

```
<ul>

 <li>River

  <ul>

   <li>Ganga</li>

   <li>Yamuna</li>

  </ul>

 </li>

 <li>Brahmaputra

  <ul style="list-style-type: circle;">

   <li>Narmada</li>

  </ul>

 </li>

</ul>
```

Q8. Types of CSS

There are three main types used for styling. Inline CSS is written directly inside an HTML tag using the style attribute. Internal CSS is written within a <style> tag in the HTML head section. External CSS is written in a separate .css file and linked to the HTML document using a <link> tag.

Q9. Practical Understanding

Think of a webpage like building a house: HTML provides the fundamental structure and skeleton (the bricks and walls). CSS acts as the interior design and paint, controlling how everything looks. JavaScript is the electricity and plumbing that makes the house interactive and functional.

Q10. Short Answer

a. What is the role of Bootstrap in web development?

Bootstrap provides ready-to-use styling classes and grid systems so developers can easily create mobile-friendly web pages without starting from scratch.

b. What is jQuery and why was it widely used in web development?
jQuery is a library that was widely used because it drastically simplified writing JavaScript for DOM manipulation and fixed cross-browser compatibility issues.

c. What is React.js and what problem does it solve in frontend development?
React.js solves the problem of updating complex UIs by using a Virtual DOM and breaking the user interface down into smaller, reusable components, making the app faster and easier to maintain.